Problem Set 2

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Due: February 18, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub in `.pdf` form.
- This problem set is due before 23:59 on Wednesday February 18, 2026. No late assignments will be accepted.

We're interested in what types of international environmental agreements or policies people support (Bechtel and Scheve 2013). So, we asked 8,500 individuals whether they support a given policy, and for each participant, we vary the (1) number of countries that participate in the international agreement and (2) sanctions for not following the agreement.

Load in the data labeled `climateSupport.RData` on GitHub, which contains an observational study of 8,500 observations.

- Response variable:
  - `choice`: 1 if the individual agreed with the policy; 0 if the individual did not support the policy
- Explanatory variables:
  - `countries`: Number of participating countries [20 of 192; 80 of 192; 160 of 192]
  - `sanctions`: Sanctions for missing emission reduction targets [None, 5%, 15%, and 20% of the monthly household costs given 2% GDP growth]

Please answer the following questions:

1. Remember, we are interested in predicting the likelihood of an individual supporting a policy based on the number of countries participating and the possible sanctions for non-compliance.

   Fit an additive model. Provide the summary output, the global null hypothesis, and p-value. Please describe the results and provide a conclusion.

Table 1:

	Dependent variable:
	choice
countries80 of 192	0.336^{***}
	(0.054)
countries160 of 192	0.648^{***}
	(0.054)
sanctions5%	0.192^{***}
	(0.062)
sanctions15%	-0.133^{**}
	(0.062)
sanctions20%	-0.304^{***}
	(0.062)
Constant	-0.273^{***}
	(0.054)
Observations	8,500
Log Likelihood	−5,784.130
Akaike Inf. Crit.	11,580.260
Note:	*p<0.1; **p<0.05; ***p<0.01

After running the global f-test we recive a pvalue of 2.2e-16 *** which at a threshold of alpha = 0.05, is an extremely signficant result. This means that at least one of our predictors can explain the variation in the model. Or that the slope of one of the predictors is different from zero.

```r
load(url("https://github.com/ASDS-TCD/StatsII_2026/blob/main/datasets/climateSupport.RData?raw=true"))

str(climateSupport)
climateSupport$countries <- factor(climateSupport$countries, ordered =
  FALSE)
climateSupport$sanctions <- factor(climateSupport$sanctions, ordered =
  FALSE)

m1 <- glm(
  choice ~ countries + sanctions,
  data = climateSupport,
  family = "binomial")

summary(m1)

stargazer(m1)

m_null <- glm(choice ~ 1, data = climateSupport, family = binomial)

gftest <- anova(m_null, m1, test = "LRT")
```

2. If any of the explanatory variables are statistically significant in this model, then:

   (a) For the policy in which nearly all countries participate [160 of 192], how does increasing sanctions from 5% to 15% change the odds that an individual will support the policy? (Interpretation of a coefficient)

   (b) On average, increasing sanctions from 5% to 15% decreases the log odds of supporting the policy by 0.18, holding all else constant. If we calucalute the odds we should take

   (c) For the policy in which very few countries participate [20 of 192], how does increasing sanctions from 5% to 15% change the odds that an individual will support the policy? (Interpretation of a coefficient)

   (d) What is the estimated probability that an individual will support a policy if there are 80 of 192 countries participating with no sanctions?

3. Would the answers to 2a and 2b potentially change if we included an interaction term in this model? Why?

```r
m_interaction <- glm(choice ~ countries * sanctions,
                     family = binomial, data = climateSupport)

anova(m1, m_interaction, test = "LRT")
```

- Perform a test to see if including an interaction is appropriate.

  After running a test with the additive model and the interactive model, we find a p-value of 0.39. At a threshold of alpha = 0.05 this is not a signifcant result, meaning that parallel lines suffice.